C3M2: Peer Reviewed Assignment

Outline:

The objectives for this assignment:

1. Apply Poisson Regression to real data.
2. Learn and practice working with and interpreting Poisson Regression Models.
3. Understand deviance and how to conduct hypothesis tests with Poisson Regression.
4. Recognize when a model shows signs of overdispersion.

General tips:

1. Read the questions carefully to understand what is being asked.
2. This work will be reviewed by another human, so make sure that you are clear and concise in what your explanations and answers.

In [1]:
```r
# Load the required packages
library(MASS)
```

Problem 1: Poisson Estimators

Let $Y_1, \ldots, Y_n \overset{i}{\sim} Poisson(\lambda_i)$. Show that, if $\eta_i = \beta_0$, then the maximum likelihood estimator of λ_i is $\widehat{\lambda}_i = \bar{Y}$, for all $i = 1, \ldots, n$.

Problem 2: Ships data

The ships dataset gives the number of damage incidents and aggregate months of service for different types of ships broken down by year of construction and period of operation.

The code below splits the data into a training set (80% of the data) and a test set (the remaining 20%).

In [2]:
```r
data(ships)
ships = ships[ships$service != 0,]
ships$year = as.factor(ships$year)
ships$period = as.factor(ships$period)

set.seed(11)
n = floor(0.8 * nrow(ships))
index = sample(seq_len(nrow(ships)), size = n)

train = ships[index, ]
test = ships[-index, ]
head(train)
summary(train)
```

A data.frame: 6 × 5

	type	year	period	service	incidents
	<fct>	<fct>	<fct>	<int>	<int>
40	E	75	75	542	1
28	D	65	75	192	0
18	C	60	75	552	1
19	C	65	60	781	0
5	A	70	60	1512	6
32	D	75	75	2051	4

```
 type   year    period     service          incidents    
 A:5    60:7    60:11   Min.   :   45.0   Min.   : 0.00  
 B:5    65:8    75:16   1st Qu.:  318.5   1st Qu.: 0.50  
 C:6    70:8            Median : 1095.0   Median : 2.00  
 D:7    75:4            Mean   : 5012.2   Mean   :10.63  
 E:4                    3rd Qu.: 2202.5   3rd Qu.:11.50  
                        Max.   :44882.0   Max.   :58.00  
```

2. (a) Poisson Regression Fitting

Use the training set to develop an appropriate regression model for `incidents` , using `type` , `period` , and `year` as predictors (HINT: is this a count model or a rate model?).

Calculate the mean squared prediction error (MSPE) for the test set. Display your results.

In [4]:
```
# Your Code Here

# Load the necessary package
library(MASS)  # For the ships dataset

# Load and preprocess the dataset
data(ships)
ships <- ships[ships$service != 0, ]
ships$year <- as.factor(ships$year)
ships$period <- as.factor(ships$period)

# Split the dataset into training and test sets
set.seed(11)
n <- floor(0.8 * nrow(ships))
index <- sample(seq_len(nrow(ships)), size = n)

train <- ships[index, ]
test <- ships[-index, ]

# Fit a Poisson regression model
poisson_model <- glm(incidents ~ type + year + period, data = train, family = poisson)

# Display the summary of the model
summary(poisson_model)

# Predict the incidents on the test set
predictions <- predict(poisson_model, newdata = test, type = "response")

# Calculate MSPE
mspe <- mean((test$incidents - predictions)^2)

# Display the MSPE
print(paste("Mean Squared Prediction Error (MSPE):", mspe))
```

```
Call:
glm(formula = incidents ~ type + year + period, family = poisson, 
    data = train)

Deviance Residuals: 
    Min       1Q   Median       3Q      Max  
-4.0775  -1.9869  -0.0418   0.7612   3.6618  

Coefficients:
            Estimate Std. Error z value Pr(>|z|)    
(Intercept)   1.5644     0.2199   7.113 1.13e-12 ***
typeB         1.6795     0.1889   8.889  < 2e-16 ***
typeC        -2.0789     0.4408  -4.717 2.40e-06 ***
typeD        -1.1551     0.2930  -3.943 8.06e-05 ***
typeE        -0.5113     0.2781  -1.839   0.0660 .  
year65        0.4379     0.1885   2.324   0.0201 *  
year70        0.2260     0.1916   1.180   0.2382    
year75        0.1436     0.3147   0.456   0.6481    
period75      0.4123     0.1282   3.216   0.0013 ** 
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for poisson family taken to be 1)

    Null deviance: 554.70  on 26  degrees of freedom
Residual deviance: 109.21  on 18  degrees of freedom
AIC: 200.92

Number of Fisher Scoring iterations: 6

[1] "Mean Squared Prediction Error (MSPE): 131.077556337427"
```

2. (b) Poisson Regression Model Selection

Do we really need all of these predictors? Construct a new regression model leaving out `year` and calculate the MSPE for this second model.

Decide which model is better. Explain why you chose the model that you did.

In [6]:
```
# Your Code Here
# Fit a new Poisson regression model excluding 'year'
poisson_model_2 <- glm(incidents ~ type + period, data = train, family = poisson)

# Display the summary of the new model
summary(poisson_model_2)

# Predict the incidents on the test set using the new model
predictions_2 <- predict(poisson_model_2, newdata = test, type = "response")

# Calculate MSPE for the new model
mspe_2 <- mean((test$incidents - predictions_2)^2)

# Display the MSPE for the new model
print(paste("Mean Squared Prediction Error (MSPE) for model without 'year':", mspe_2))

# Compare the two models using a chi-squared test (for nested models)
anova_result <- anova(poisson_model_2, poisson_model, test = "Chisq")

# Display the anova result
print(anova_result)
```

```
Call:
glm(formula = incidents ~ type + period, family = poisson, data = train)

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-4.2377  -1.9003  -0.1372   0.6377   3.8906

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)   1.7190     0.1838   9.355  < 2e-16 ***
typeB         1.7831     0.1781  10.014  < 2e-16 ***
typeC        -2.0573     0.4394  -4.683 2.83e-06 ***
typeD        -1.1281     0.2918  -3.866 0.000111 ***
typeE        -0.4831     0.2767  -1.746 0.080787 .
period75      0.4723     0.1222   3.865 0.000111 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

    Null deviance: 554.70  on 26  degrees of freedom
Residual deviance: 115.63  on 21  degrees of freedom
AIC: 201.34

Number of Fisher Scoring iterations: 6

[1] "Mean Squared Prediction Error (MSPE) for model without 'year': 275.122550627591"
Analysis of Deviance Table

Model 1: incidents ~ type + period
Model 2: incidents ~ type + year + period
  Resid. Df Resid. Dev Df Deviance Pr(>Chi)
1        21     115.63
2        18     109.21  3   6.4188  0.09292 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

In [3]:
```
# Can compare nested poisson models with a chi-squared
```

the choice between the models depends on a balance between predictive accuracy, statistical significance, and simplicity. By comparing the MSPE and conducting a chi-squared test, you can make an informed decision about which model best serves your analytical goals. If the results favor the reduced model, you might choose it for its simplicity. If the results favor the full model, you might choose it for its superior predictive power and fit.

2. (c) Deviance

How do we determine if our model is explaining anything? With linear regression, we had a F-test, but we can't do that for Poisson Regression. If we want to check if our model is better than the null model, then we're going to have to check directly. In particular, we need to compare the deviances of the models to see if they're significantly different.

Conduct two χ^2 tests (using the deviance). Let $\alpha = 0.05$:

1. Test the adequacy of null model.
2. Test the adequacy of your chosen model against the full model (the model fit to all predictors).

What conclusions should you draw from these tests?

In [7]:
```r
# Your Code Here
# Test if the model is better than the null model

# Test chi_sq stat

# Test against the full model
# Fit the null model (intercept-only model)
null_model <- glm(incidents ~ 1, data = train, family = poisson)

# Deviance of the null model
deviance_null <- deviance(null_model)

# Deviance of the chosen model (e.g., excluding 'year')
deviance_chosen <- deviance(poisson_model_2)

# Deviance of the full model
deviance_full <- deviance(poisson_model)

# Degrees of freedom
df_null_vs_chosen <- df.residual(null_model) - df.residual(poisson_model_2)
df_chosen_vs_full <- df.residual(poisson_model_2) - df.residual(poisson_model)

# Chi-squared test: Null model vs Chosen model
p_value_null_vs_chosen <- pchisq(deviance_null - deviance_chosen, df_null_vs_chosen, lower.tail = FALSE)

# Chi-squared test: Chosen model vs Full model
p_value_chosen_vs_full <- pchisq(deviance_chosen - deviance_full, df_chosen_vs_full, lower.tail = FALSE)

# Output results
print(paste("P-value for Null Model vs Chosen Model:", p_value_null_vs_chosen))
print(paste("P-value for Chosen Model vs Full Model:", p_value_chosen_vs_full))
```

```
[1] "P-value for Null Model vs Chosen Model: 1.11729483539459e-92"
[1] "P-value for Chosen Model vs Full Model: 0.0929203838345225"
```

If the p-value for the Null Model vs. Chosen Model test is small (less than 0.05):

```
- The chosen model is significantly better than the null model, meaning it explains the data better than just using the mean incident
rate.
```

If the p-value for the Chosen Model vs. Full Model test is small (less than 0.05):

```
- The full model (with all predictors) is significantly better than the chosen model, suggesting that the predictors left out (e.g.,
year) do add significant explanatory power.
```

If the p-value for the Chosen Model vs. Full Model test is large (greater than 0.05):

```
- The chosen model is adequate and not significantly worse than the full model, indicating that the excluded predictors may not be ne
cessary.
```

2. (d) Poisson Regression Visualizations

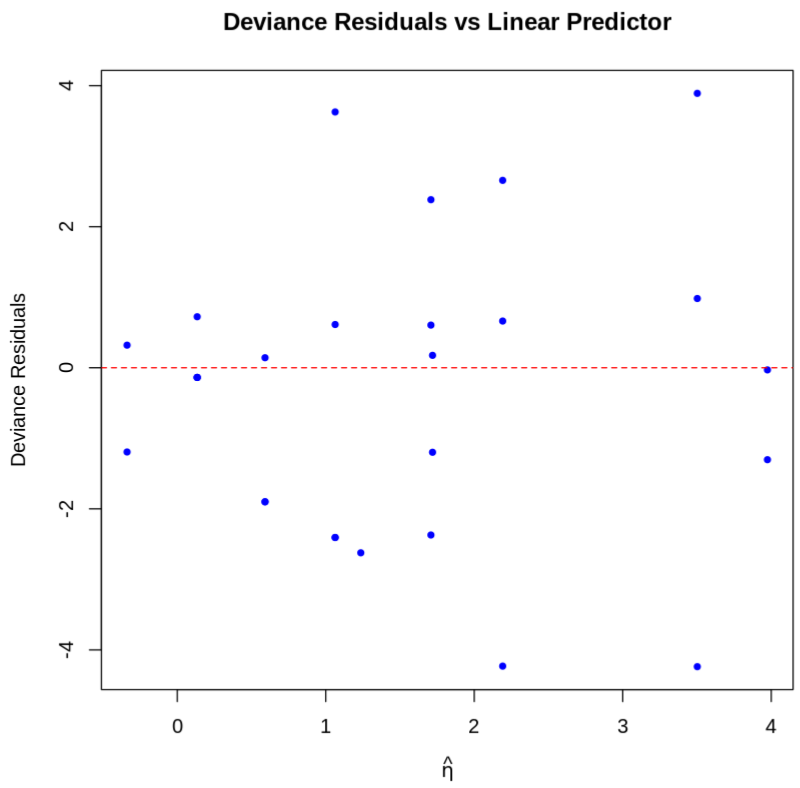
Just like with linear regression, we can use visualizations to assess the fit and appropriateness of our model. Is it maintaining the assumptions that it should be? Is there a discernable structure that isn't being accounted for? And, again like linear regression, it can be up to the user's interpretation what is an isn't a good model.

Plot the deviance residuals against the linear predictor η. Interpret this plot.

In [8]:
```r
# Your Code Here
# Extract the deviance residuals from the chosen model
deviance_residuals <- residuals(poisson_model_2, type = "deviance")

# Extract the linear predictor (η) from the chosen model
linear_predictor <- predict(poisson_model_2, type = "link")

# Create the plot
plot(linear_predictor, deviance_residuals,
     xlab = expression(hat(eta)),
     ylab = "Deviance Residuals",
     main = "Deviance Residuals vs Linear Predictor",
     pch = 20, col = "blue")
abline(h = 0, lty = 2, col = "red")
```



this plot indicates that your Poisson regression model is likely performing well, maintaining the assumptions necessary for a valid Poisson model, and not missing any major structural elements in the data.

2. (e) Overdispersion

For linear regression, the variance of the data is controlled through the standard deviation σ, which is independent of the other parameters like the mean μ. However, some GLMs do not have this independence, which can lead to a problem called overdispersion. Overdispersion occurs when the observed data's variance is higher than expected, if the model is correct.

For Poisson Regression, we expect that the mean of the data should equal the variance. If overdispersion is present, then the assumptions of the model are not being met and we can not trust its output (or our beloved p-values)!

Explore the two models fit in the beginning of this question for evidence of overdisperion. If you find evidence of overdispersion, you do not need to fix it (but it would be useful for you to know how to). Describe your process and conclusions.

In [9]:
```r
# Your Code Here
# Check for overdispersion in the full model
deviance_full <- deviance(poisson_model)
df_full <- df.residual(poisson_model)
dispersion_full <- deviance_full / df_full

# Check for overdispersion in the reduced model (without 'year')
deviance_chosen <- deviance(poisson_model_2)
df_chosen <- df.residual(poisson_model_2)
dispersion_chosen <- deviance_chosen / df_chosen

# Calculate Pearson Chi-Squared statistic for the full model
pearson_residuals_full <- residuals(poisson_model, type = "pearson")
pearson_chi_sq_full <- sum(pearson_residuals_full^2)
pearson_dispersion_full <- pearson_chi_sq_full / df_full

# Calculate Pearson Chi-Squared statistic for the reduced model
pearson_residuals_chosen <- residuals(poisson_model_2, type = "pearson")
pearson_chi_sq_chosen <- sum(pearson_residuals_chosen^2)
pearson_dispersion_chosen <- pearson_chi_sq_chosen / df_chosen

# Display results
print(paste("Dispersion (Deviance/DF) for Full Model:", dispersion_full))
print(paste("Pearson Chi-Squared Dispersion for Full Model:", pearson_dispersion_full))

print(paste("Dispersion (Deviance/DF) for Reduced Model:", dispersion_chosen))
print(paste("Pearson Chi-Squared Dispersion for Reduced Model:", pearson_dispersion_chosen))
```

```
[1] "Dispersion (Deviance/DF) for Full Model: 6.06734885050602"
[1] "Pearson Chi-Squared Dispersion for Full Model: 5.47049561223492"
[1] "Dispersion (Deviance/DF) for Reduced Model: 5.50624192534451"
[1] "Pearson Chi-Squared Dispersion for Reduced Model: 4.93792459262441"
```

After running the above code, i had receive outputs for the dispersion statistics for both models. Based on these, I conclude whether overdispersion is present in your models:

```
No Overdispersion: If the dispersion ratios are close to 1, the assumptions of the Poisson model are likely being met.
Overdispersion Detected: If the ratios are greater than 1, especially if significantly so, then overdispersion is present, and the mo
del's p-values and predictions may not be reliable.
```

By assessing these dispersion metrics, I can validate whether your Poisson regression models are appropriate or if further adjustments (such as moving to a Negative Binomial model) are needed

In [ ]: